Process Report: VU-DM-2026-Group-5

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Workflow and discipline

We worked through nine model-development stages, beginning with a bootstrap and EDA stage and progressing through anchor modelling, feature-hypothesis testing, model comparison and bias mitigation, ensemble extensions, the historical-prior model, and a final exploratory stage. Each stage changed one main element on the main branch, never two, and was preceded by a short written plan that listed scope, hypotheses, expected impact, and an acceptance criterion. Acceptance was decided by same-seed deltas on a locked holdout (sha256 prefix `bc8ea6f6`, 19,980 `srch_id` queries, group-disjoint from train and validation), not by raw means against a moving anchor. Train-only columns were dropped before features were built; target encoding used 5-fold group K-fold on `srch_id`, never train-test joint statistics. We pinned thread counts (`num_threads=6`) and never used `-1`. Memory peaked under 10 GB during training; the working laptop was usable throughout.

Timeline

- **Bootstrap and EDA stage (2026-05-06).** EDA on full data plus a five percent stratified sample, locked holdout JSON with sha256 `bc8ea6f6`, four heuristic baselines on the locked holdout (random 0.1579, star 0.1949, price 0.1987, location score 2 0.2836).
- **LambdaMART anchor stage (2026-05-06).** LightGBM LambdaMART on the 49 raw-feature anchor, three seeds, mean holdout NDCG@5 = 0.392667 with std 0.001070, three-seed score-average 0.395289.
- **Feature-hypothesis stage (2026-05-06).** Five hypothesis blocks: per-search normalisation features, leave-one-out property priors, a mean-position proxy, negative downsampling, and competitor and price-composite features. All five rejected on same-seed delta against the anchor.
- **Model comparison and bias stage (2026-05-06 to 2026-05-07).** Hyperparameter exploration (label-gain variants, depth, learning rate) rejected; documentation check confirmed that `lambdarank_unbiased` is not a parameter in LightGBM 4.x. XGBoost `rank:ndcg` compared as a sample-only second technique (LightGBM 0.329601 versus XGBoost 0.356409 on the 5 % sample). Bias detection by visitor country and by property popularity tertile; post-processing exposure rerank with $\lambda = 0.20$ (global NDCG@5 drop 0.003090; medium and low tertile exposure rise by 5.3 and 2.2 percentage points respectively). Safe-baseline checkpoint and Kaggle upload (`submit_anchor_3seed_avg.csv`, public 0.39500), version-control checkpoint `final_safe_baseline_public_039500`.
- **Ensemble extension stage (2026-05-07).** Five-seed extension (seeds 42, 123, 456, 789, 2026) holdout 0.397348; with a post-hoc additive blend over per-search z-scored signals holdout 0.398246. Kept as a local result, not uploaded.
- **Diverse-ensemble stage (2026-05-07).** Diverse-config LightGBM (depth 8, 63 leaves, learning rate 0.035, $\lambda_2 = 10$, bag 0.85, feature 0.75) over 3 seeds, weighted 0.60 anchor plus 0.40 diverse with the same blend, holdout 0.399638. Manual Kaggle upload, public 0.39685.
- **Historical-prior stage (2026-05-07).** Twelve leakage-safe historical priors over `prop_id`, `srch_destination_id`, and the pair, with K-fold encoding. Three-seed score-average holdout 0.400682 (first local result above 0.4). Manual Kaggle upload, public 0.39690 (final selected submission).
- **Final exploratory stage (2026-05-07).** A pre-specified rank blend over weight sets W0–W6 reached holdout 0.401482 at its maximum but we did not select it; the conservative pick at 0.400453 was also not uploaded. A fourth prior block keyed by `(prop_id, visitor_location_country_id)` gave a same-seed delta of +0.000745, below the +0.0010 extension threshold, and was rejected.
- **Finalisation stage (2026-05-07).** Clean folder `Assignment 2/` produced from the working folder: structure plus notebook outlines; figure generation from cached parquet and earlier report card JSON; scientific and process report drafts; verified Bellucci 2021 citation; final read-through and submission package.

Critical reflection

The most important lesson is the local-to-public shrinkage we observed: the public-versus-local gap moved from +0.000289 (safe baseline) to −0.002788 (diverse ensemble) to −0.003782 (final submission). Each individual decision was leakage-safe, but the cumulative search budget on a fixed locked holdout inflated expected public gains. The conservative choice in the final exploratory stage (not uploading the holdout-max blend, keeping the simpler historical-prior 3-seed model as the headline) is our explicit response to this: we treated holdout NDCG@5 as a selection signal that loses calibration over many stages and preferred the candidate whose simplicity reduced the reasoning chain.

A second lesson is the importance of feature provenance: every feature in the final pipeline traces back to a primary source or to a verified leakage-safety argument, and each rejected feature has an experiment record documenting the same-seed delta. This made it cheap to discard hypotheses without lingering doubt.

Tooling and process

Documentation. Short Markdown files in the working folder track state across sessions: current status, prioritised work, accept/reject decisions, durable lessons, pitfalls, report-worthy claims, and a session-by-session timeline that this report draws on. Each stage also appends a row to a structured CSV log of experiments, and each public leaderboard upload is recorded on the same row.

Version-control checkpoints. Each accepted milestone is captured by a git commit and tag, beginning with a checkpoint at the safe baseline submission, followed by a local-only checkpoint after the five-seed blend, a checkpoint after the diverse ensemble submission, and a checkpoint after the historical-prior submission. We preserved versioned outputs (the three uploaded submission CSVs and the relevant per-stage JSON summaries) by sha256 between sessions and re-verified them at the start and end of each working session.

Leaderboard usage. We treated the public leaderboard strictly as an external check, not as a tuning signal. We uploaded three submissions in total over the project (safe baseline, diverse ensemble, historical-prior submission). The final exploratory stage produced two candidate CSVs that we deliberately did not upload, both because the holdout-max selection was the wrong move under our shrinkage observation and because public-leaderboard tuning was outside our chosen protocol.

Task division

The work was a collaborative effort and most steps were cross-reviewed. The lead-contributor split below reflects who drove each thread day to day.

- **Hidde Franke**: project orchestration, the staged development plan, the LightGBM modelling track from the LambdaMART anchor stage through the historical-prior stage, the leakage-safe prior encoding, ensembling, the three Kaggle uploads, the clean folder and `src/` pipeline, and report assembly.
- **Hendrico Burger**: EDA support and figure inspection, feature-engineering review and ablation sanity checks within the feature-hypothesis stage, validation of same-seed deltas and the locked holdout, and process documentation across the development stages.
- **Joanne Pals**: the bias and deployment thread within the model comparison and bias stage, fairness interpretation including the popularity-tertile rerank with $\lambda = 0.20$, scientific and process report editing, and process report review.

Reproducibility

The clean folder `Assignment 2/` contains the deliverable code under `src/`, the explanatory notebooks under `notebooks/`, and the final submission CSV in `results/`. The locked holdout JSON, its sha256 hash, and the per-seed best-iteration values are in `artifacts/`. The command `python src/pipeline.py --train --predict` reproduces the final submission from the processed parquet files; thread counts are pinned and we never use −1. `python src/pipeline.py --validate results/submit_iter07_prior_3seed.csv` validates the included submission CSV without retraining.